

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology)

Ph.D. in Artificial Intelligence

Advisor: [Prof. Jinwoo Shin](#)

Sep 2024 – Present

Daejeon, South Korea

KAIST (Korea Advanced Institute of Science and Technology)

B.S. in Computer Science (School of Computing)

Advisor: [Prof. Sung-Eui Yoon](#)

Mar 2020 – Aug 2024

Daejeon, South Korea

RESEARCH INTERESTS

My goal is to build generally intelligent robots that can understand and adapt to diverse real-world situations. To achieve this, my research focuses on Vision–Language–Action (VLA) models and world models. I believe that grounding rich world understanding through these models is key to building general-purpose robot policies. To this end, I have worked on aligning VLMs and LLMs with reinforcement learning, improving the spatiotemporal understanding of VLMs, learning world models for robot policy augmentation.

Keywords: VLA, World model, VLM, RL, Robotics

PUBLICATIONS (* denotes equal contribution, § denotes research lead)

[T1] [RLDX-1 Technical Report](#)

Dongyoung Kim[§], Huiwon Jang[§], Myungkyu Koo*, Suhyeok Jang*, Taeyoung Kim*, Beomjun Kim, Byungjun Yoon, Changsung Jang, Daewon Choi, Dongsu Han, Donguk Lee, Heeseung Kwon, Hojin Jeon, Jaehyun Kang, Jaekyoung Bae, Jihyuk Lee, Jimin Lee, John Won, Joonwoo Ahn, Junhyeong Park, Junyoung Sung, Kyungmin Lee, Minseong Han, Minsung Yoon, Sejune Joo, Seonil Son, Seungcheol Park, Seunggeun Cho, Seungjun Moon, Seungku Kim, Yonghoon Dong, Yongjin Cho, Youngchan Kim, Chang Hwan Kim, Dohyeon Kim, Heecheol Kim, Heewon Lee, Hensen Ahn, Hyungkyu Ryu, Hyunsoo Choi, Hyunsoo Shin, Jaehoon Jung, Jaewoo Kim, Jinwook Kim, Joochul Chang, Joonsoo Kim, Junghun Park, Jungwoo Park, Junho Cho, Junhyeok Park, Junwon Lee, Kangwook Lee, Kwanghoon Kim, Kyoungwhan Choe, Manoj Bhadu, Nayoung Oh, Sangjun Kim, Sangwoo Kim, Seunghoon Shim, Seunghyun Kim, Seungjun Lee, Seungyup Ka, Sungryol Yang, Wook Jung, Yashu Shukla, Yeonjae Lee, Yeonwoo Bae, [Jinwoo Shin](#)[§]
Technical Report, 2026

[C9] [Dual-Stream Diffusion for World-Model Augmented Vision-Language-Action Model](#)

John Won, Kyungmin Lee, [Huiwon Jang](#), **Dongyoung Kim**, [Jinwoo Shin](#)
International Conference on Machine Learning (ICML), 2026

[C8] [Contrastive Representation Regularization for Vision-Language-Action Models](#)

Taeyoung Kim, Jimin Lee, Myungkyu Koo, **Dongyoung Kim**, Kyungmin Lee, Changyeon Kim, [Younggyo Seo](#), [Jinwoo Shin](#)
International Conference on Machine Learning (ICML), 2026

[C7] [Verifier-free Test-Time Sampling for Vision Language Action Models](#)

Suhyeok Jang, **Dongyoung Kim**, Changyeon Kim, Youngsuk Kim, [Jinwoo Shin](#)
International Conference on Learning Representations (ICLR), 2026

[C6] [Robot-R1: Reinforcement Learning for Enhanced Embodied Reasoning in Robotics](#)

Dongyoung Kim, Sumin Park, [Huiwon Jang](#), [Jinwoo Shin](#), [Jaehyung Kim](#)*, [Younggyo Seo](#)*
Neural Information Processing Systems (NeurIPS), 2025

- [C5] Debiasing Online Preference Learning via Preference Feature Preservation
Dongyoung Kim, Jinsung Yoon, Jinwoo Shin, Jaehyung Kim
Annual Meeting of the Association for Computational Linguistics (ACL), 2025 (Findings)
- [C4] Spread Preference Annotation: Direct Preference Judgment for Efficient LLM Alignment
Dongyoung Kim, Kimin Lee, Jinwoo Shin, Jaehyung Kim
International Conference on Learning Representations (ICLR), 2025
Oral Presentation (207/11672=1.77%)
- [C3] Learning to Correct for QA Reasoning with Black-box LLMs
 Jaehyung Kim, **Dongyoung Kim**, Yiming Yang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024 (Main)
- [C2] Visual Representation Learning with Stochastic Frame Prediction
 Huiwon Jang, **Dongyoung Kim**, Junsu Kim, Jinwoo Shin, Pieter Abbeel, Younggyo Seo
International Conference on Machine Learning (ICML), 2024
- [C1] Accelerating Reinforcement Learning with Value-Conditional State Entropy Exploration
Dongyoung Kim, Jinwoo Shin, Pieter Abbeel, Younggyo Seo
Neural Information Processing Systems (NeurIPS), 2023
- [W1] Training-free LLM Verification via Recycling Few-shot Examples
 Dongseok Lee, Jimyung Hong, **Dongyoung Kim**, Jaehyung Kim
International Conference on Machine Learning (ICML) Workshop ES-FoMo-III, 2025
- [P5] RoboAlign: Reinforcement Learning for Action-Aligned Multimodal Large Language Models
Dongyoung Kim, Sumin Park, Woomin Song, Seungku Kim, Taeyoung Kim, Huiwon Jang, Jaehyung Kim, Younggyo Seo, Jinwoo Shin
 Preprint, 2025
- [P4] RoboCurate: Harnessing Diversity with Action-Verified Neural Trajectory for Robot Learning
 Seungku Kim*, Suhyeok Jang*, Byungjun Yoon, **Dongyoung Kim**, John Won, Jinwoo Shin
 Preprint, 2025
- [P3] INFUSER: Injecting Synthetic Failures for Self-Correcting Embodied Agents
 Byungjun Yoon, **Dongyoung Kim**, Jaehyun Nam, Wook Jung, Jinwoo Shin
 Preprint, 2025
- [P2] Collaborative LLM Inference via Planning for Efficient Reasoning
 Byeongchan Lee, Jonghoon Lee, **Dongyoung Kim**, Jaehyung Kim, Jinwoo Shin
 Preprint, 2025
- [P1] SpatialBoost: Enhancing Visual Representation through Language-Guided Reasoning
 Byoungwooo Jeon*, **Dongyoung Kim***, Huiwon Jang, Jinwoo Shin
 Preprint, 2025

HONORS AND AWARDS

- Recipient, Google Conference Scholarships (APAC), May 2025
- Travel Award, Conference on Neural Information Processing Systems (NeurIPS), 2023
- Recipient, KAIST Presidential Fellowship, 2020 – 2024

WORK EXPERIENCE

RLWRLD Inc.

June 2025 – Present

Seoul, South Korea

Research Intern (Advisor: Jinwoo Shin)

Reviewer

- ICML
- NeurIPS
- ICLR